



LUND
UNIVERSITY

Written Examination
Linear Analysis, MATB24
Formula sheet

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Power series

1. The radius of convergence R of a power series $\sum_{n=0}^{\infty} a_n(x-a)^n$ obeys $1/R = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}$ and (if the limit exists) $1/R = \lim_{n \rightarrow \infty} |a_{n+1}|/|a_n|$.
2. $e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$.
3. $\sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!} = \frac{1}{2i}(e^{iz} - e^{-iz})$.
4. $\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!} = \frac{1}{2}(e^{iz} + e^{-iz})$.
5. $\sinh z = \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!} = \frac{1}{2}(e^z - e^{-z})$.
6. $\cosh z = \sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!} = \frac{1}{2}(e^z + e^{-z})$.
7. $e^{iz} = \cos z + i \sin z$.
8. $e^z = \cosh z + \sinh z$.

Convolution

9. $(f * g)(t) = \int_I f(t-s)g(s) ds$, for $f, g \in L^1(I)$, $I \in \{\mathbb{T}, \mathbb{R}\}$.

Fourier series

Let $f \in L^1(\mathbb{T})$.

10. $f(t) \sim \sum_{n \in \mathbb{Z}} \hat{f}(n)e^{int}$, where $\hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t)e^{-int} dt$.
11. $f(t) \sim a_0/2 + \sum_{n=1}^{\infty} (a_n \cos(nt) + b_n \sin(nt))$, where $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(nt) dt$ and $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt) dt$.
12. $\widehat{e^{ia \cdot} f}(n) = \hat{f}(n-a)$, for $a \in \mathbb{Z}$.
13. $\widehat{f(\cdot - b)}(n) = e^{-inb} \hat{f}(n)$, for $b \in \mathbb{R}$.
14. $\widehat{f * g}(n) = 2\pi \hat{f}(n) \hat{g}(n)$, for $g \in L^1(\mathbb{T})$.
15. $\widehat{f'}(n) = in \hat{f}(n)$, if $f \in C^1(\mathbb{T})$.
16. $\int_{-\pi}^{\pi} |f(t)|^2 dt = 2\pi \sum_{n \in \mathbb{Z}} |\hat{f}(n)|^2$, if $f \in L^2(\mathbb{T})$.

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17. $\int_{-\pi}^{\pi} |f(t)|^2 dt = \pi (|a_0|^2/2 + \sum_{n=1}^{\infty} (|a_n|^2 + |b_n|^2)),$ if $f \in L^2(\mathbb{T})$.

18. $\int_{-\pi}^{\pi} f(t) \overline{g(t)} dt = 2\pi \sum_{n \in \mathbb{Z}} \hat{f}(n) \overline{\hat{g}(n)},$ if $f, g \in L^2(\mathbb{T})$.

Let $f \in L^1((0, \pi))$.

19. $f(t) \sim a_0/2 + \sum_{n=1}^{\infty} a_n \cos(nt),$ where $a_n = \frac{2}{\pi} \int_0^{\pi} f(t) \cos(nt) dt$.

20. $f(t) \sim \sum_{n=1}^{\infty} b_n \sin(nt),$ where $b_n = \frac{2}{\pi} \int_0^{\pi} f(t) \sin(nt) dt$.

Fourier transform

Let $f \in L^1(\mathbb{R})$.

21. $\hat{f}(\omega) = \int_{\mathbb{R}} f(t) e^{-i\omega t} dt.$

22. $f(t) = \frac{1}{2\pi} \widehat{\hat{f}}(-t) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\omega) e^{i\omega t} d\omega,$ under certain assumptions on $f, \hat{f}$.

23. $\widehat{e^{ia \cdot} f}(\omega) = \hat{f}(\omega - a),$ for $a \in \mathbb{R}$.

24. $\widehat{f(\cdot - a)}(\omega) = e^{-i\omega a} \hat{f}(\omega),$ for $a \in \mathbb{R}$.

25. $\widehat{f(a \cdot)}(\omega) = \frac{1}{|a|} \hat{f}(\omega/a),$ for $a \in \mathbb{R} \setminus \{0\}$.

26. $\widehat{f * g}(\omega) = \hat{f}(\omega) \hat{g}(\omega),$ for $g \in L^1(\mathbb{T})$.

27. $\widehat{f'}(\omega) = i\omega \hat{f}(\omega),$ if f is differentiable with $f' \in L^1(\mathbb{R})$.

28. $\widehat{t \mapsto t f(t)}(\omega) = i(\hat{f})'(\omega),$ if $(t \mapsto t f(t)) \in L^1(\mathbb{R})$.

29. $\int_{\mathbb{R}} |f(t)|^2 dt = \frac{1}{2\pi} \int_{\mathbb{R}} |\hat{f}(\omega)|^2 d\omega,$ if $f \in L^2(\mathbb{R})$.

30. $\int_{\mathbb{R}} f(t) \overline{g(t)} dt = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\omega) \overline{\hat{g}(\omega)} d\omega,$ if $f, g \in L^2(\mathbb{R})$.